

1. Questions

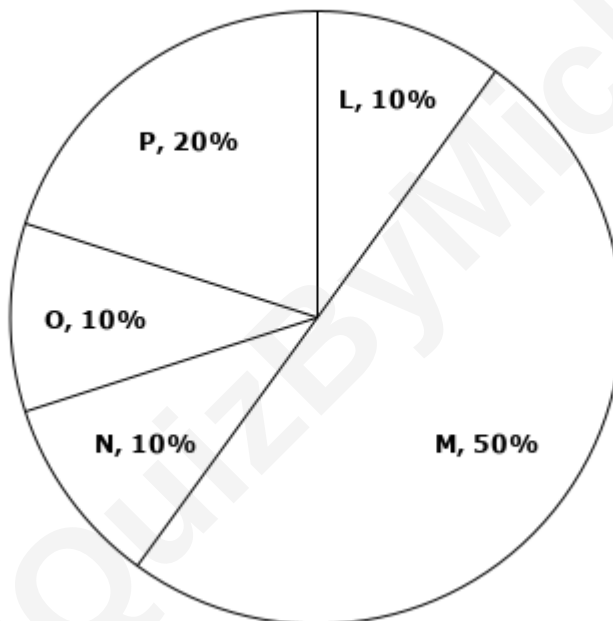
Study the following information carefully and answer the questions given below.

Two pie charts are given below. One of the pie charts represents the percentage distribution of the number of Petrol cars manufactured by five different companies L, M, N, O and P. The other pie chart represents the percentage distribution of the total number of cars (Petrol + Diesel) manufactured by these five companies.

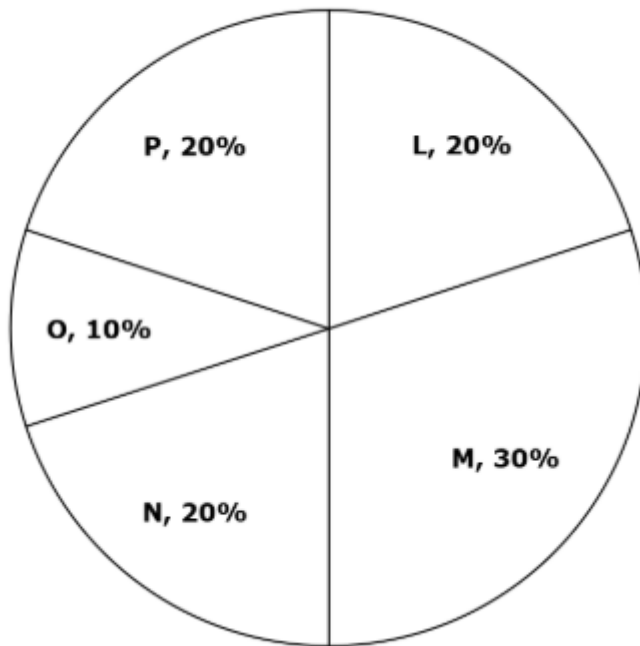
Notes:

- 1). The total number of Petrol cars manufactured by company O is 120.
- 2). The total number of cars (Petrol + Diesel) manufactured by company M is 300 more than the total number of cars (Petrol + Diesel) manufactured by company P.

% distribution of the total number of Petrol cars manufactured



% distribution of the total number of cars (Petrol + Diesel) manufactured



The total number of Diesel cars manufactured by companies L and M together is what percentage more than the number of Diesel cars manufactured by company N?

- a. 55.5%
- b. 62.5%
- c. 66.66%
- d. 58.25%
- e. 60%

2. Questions

In company A, the number of Petrol cars manufactured is 40% more than that of company M, and the number of Diesel cars manufactured is 20% less than that of company O. Find the total number of cars manufactured by company A.

- a. 964
- b. 972
- c. 984
- d. 996
- e. 1008

3. Questions

Find the total number of Petrol cars manufactured by companies L, M, and N together.

- a. 900

- b. 800
- c. 600
- d. 750
- e. 840

4. Questions

Find the ratio of the total number of Petrol cars manufactured by companies M, N and O together to the total number of Diesel cars manufactured by companies L, O and P together.

- a. 12 : 15
- b. 13 : 16
- c. 14 : 17
- d. 15 : 19
- e. 16 : 21

5. Questions

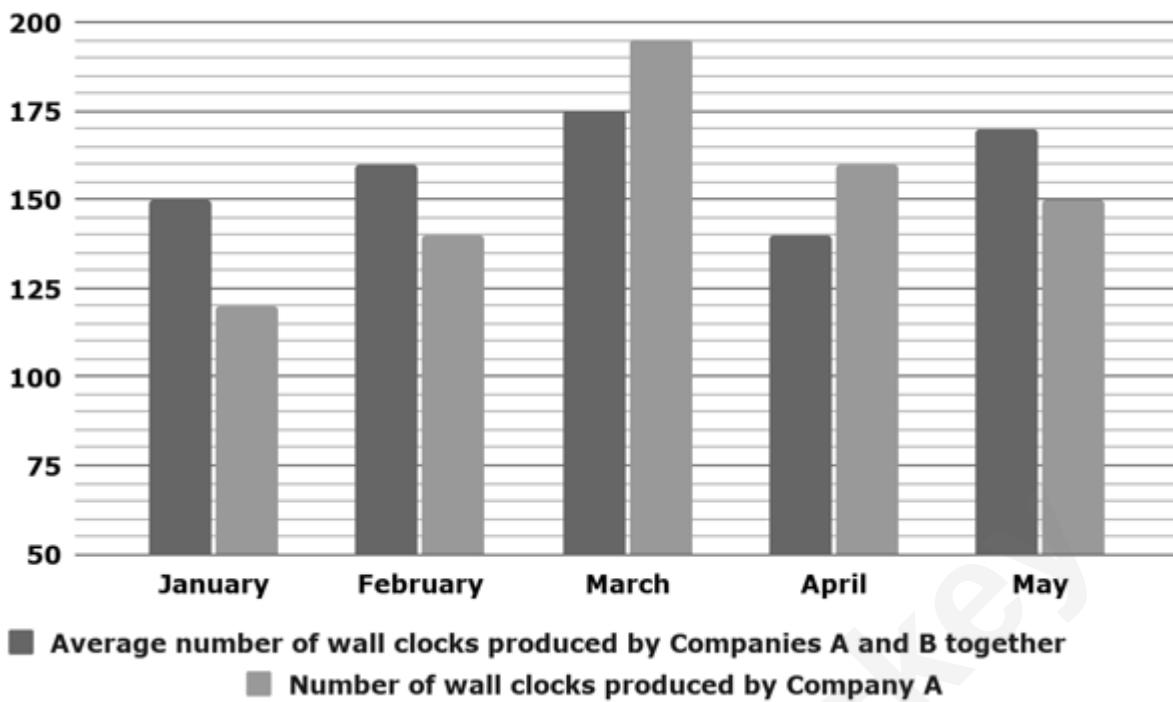
The number of Diesel cars manufactured by company M is what percentage of the total number of cars manufactured by company M?

- a. 30%
- b. 33.33%
- c. 35.5%
- d. 37.5%
- e. 40%

6. Questions

Study the following information carefully and answer the questions given below.

The bar graph given below shows the average number of wall clocks produced by Company A and Company B together and the number of wall clocks produced by Company A across five different months of the year 2026.



Note: The number of wall clocks produced by Company B = The total number of wall clocks produced by both companies together – The number of wall clocks produced by Company A.

Find the ratio of the total number of wall clocks produced by Companies A and B together in January to the total number of wall clocks produced by Companies A and B together in April.

- a. 13 : 12
- b. 15 : 14
- c. 16 : 15
- d. 14 : 13
- e. 17 : 15

7. Questions

If the total number of wall clocks produced by Companies A and B together in June was 50% more than that in April, and the ratio of the number of wall clocks produced by Company A to Company B in June was 5 : 2, then find the number of wall clocks produced by Company B in June.

- a. 100
- b. 160
- c. 140
- d. 120
- e. 180

8. Questions

The total number of wall clocks produced by both companies together in May is what percentage

more than the total number of wall clocks produced by both companies together in February?

- a. 5.25%
- b. 6.25%
- c. 7.5%
- d. 4.75%
- e. 8.5%

9. Questions

In the month of June, the number of wall clocks produced by Company A was 30% more than its own production in May, and the number of wall clocks produced by Company B in June was 20% more than its own production in April. What is the total number of wall clocks produced by both companies together in June?

- a. 339
- b. 333
- c. 325
- d. 345
- e. 351

10. Questions

Find the total number of wall clocks produced by Company A from January to April.

- a. 595
- b. 605
- c. 615
- d. 625
- e. 585

11. Questions

A vessel is filled with 250 litres of liquid A. From it, 25 litres is drawn out and replaced with liquid B, and this operation is carried out two more times. Find the quantity of liquid A remaining in the vessel.

- a. 168.5 litres
- b. 175.25 litres
- c. 182.25 litres
- d. 190.75 litres

e. 196.5 litres

12. Questions

The present ages of Kesav and Kumar are in the ratio 3 : 5, and the present ages of Kesav and Kamal are in the ratio 6 : 7. If the average of the present ages of Kumar and Kamal is 34 years, find the age of Kesav after 5 years.

- a. 26 years
- b. 32 years
- c. 27 years
- d. 29 years
- e. 31 years

13. Questions

A train covers one-sixth of the total distance in the first stage and half of the remaining distance in the second stage. The remaining distance is covered in the third and fourth stages in the ratio 4:1, respectively. If the distance covered in the second stage is 120 km more than that covered in the fourth stage, find the time taken to cover the first stage at a speed of 30 km/hr.

- a. 1 hour
- b. 3 hours
- c. 2 hours
- d. 4 hours
- e. 5 hours

14. Questions

A bag contains 8 red balls, 7 yellow balls and 9 blue balls. If one ball is drawn at random, find the probability that it is either a red ball or a yellow ball.

- a. $\frac{5}{8}$
- b. $\frac{3}{8}$
- c. $\frac{7}{12}$
- d. $\frac{2}{3}$
- e. $\frac{5}{12}$

15. Questions

A shopkeeper sells an article for Rs.1584 after allowing a discount of 12% and still earns a profit of 20%. If he sells the article at the marked price without any discount, find his profit percentage.

- a. 36.36%

- b. 33.33%
- c. 32.32%
- d. 38.38%
- e. 40.40%

16. Questions

P completes one-fourth of a work in 3 days, Q one-third of it in 6 days, and R half of it in 18 days. All three work together for n days, then P and R leave and Q alone finishes the rest in 6 days. Find the value of n.

- a. 3
- b. 5
- c. 6
- d. 4
- e. 8

17. Questions

A certain sum invested at compound interest grows to Rs.10000 at the end of 2 years and to Rs.12500 at the end of 3 years. Find the sum invested.

- a. Rs.5800
- b. Rs.6400
- c. Rs.6000
- d. Rs.7200
- e. Rs.6800

18. Questions

The sum of the perimeter of a semicircle and the perimeter of a square is 280 cm, and the ratio of the side of the square to the radius of the semicircle is 5 : 7. Find the area of the semicircle.

- a. 1540 cm²
- b. 1732 cm²
- c. 1848 cm²
- d. 1925 cm²
- e. 2156 cm²

19. Questions

The speed of a boat in still water is 13 km/hr. If the boat covers 60 km upstream and 96 km downstream in a total of 12 hours, find the time taken by the boat to cover 45 km upstream.

- a. 3 hours 30 minutes
- b. 5 hours
- c. 4 hours
- d. 4 hours 30 minutes
- e. 5 hours 30 minutes

20. Questions

A and B started a business by investing Rs.3600 and Rs.4800 respectively. After 4 months, B withdrew Rs.1200. If the total profit at the end of the year was Rs.28500, find the share of B.

- a. Rs.13500
- b. Rs.12000
- c. Rs.15000
- d. Rs.16500
- e. Rs.14250

21. Questions

What value will come in place of the question mark (?) in the following question?

$$16 * 8 + 40\% \text{ of } 1500 = ? + 68$$

- a. 640
- b. 680
- c. 660
- d. 620
- e. 650

22. Questions

$$47 + 15 * \frac{6}{25} - 12.6 = ? * \frac{16}{4}$$

- a. 8.5
- b. 9.5
- c. 9.2

d. 10.5

e. 8.8

23. Questions

$$\sqrt{[12 * 12 \div (1/9)]} + 18 * 15 = ? * 18$$

a. 15

b. 19

c. 17

d. 21

e. 14

24. Questions

$$35 * \frac{24}{56} + \frac{84}{3} - \frac{77}{5} = ?$$

a. 26.4

b. 28.2

c. 28.6

d. 29.4

e. 27.6

25. Questions

$$? \div 5 + 144 \div 2.4 * 2 = 14 * 15$$

a. 450

b. 480

c. 420

d. 400

e. 460

26. Questions

What approximate value will come in place of the question mark (?) in the following question?

$$\sqrt{399.99 * 17.99 \div 5.98} = ?^2 + (63.99)^{1/3} * 6.01$$

a. 36

b. 12

- c. 6
- d. 60
- e. 24

27. Questions

$$149.99\% \text{ of } 199.98 - 359.99 \div \sqrt{143.99} = \sqrt{323.99} * ?$$

- a. 5
- b. 30
- c. 20
- d. 15
- e. 10

28. Questions

$$(24.99)^2 + (19.98)^2 - (29.99)^2 + \sqrt{255.99} = ?$$

- a. 141
- b. 148
- c. 135
- d. 152
- e. 129

29. Questions

$$79.98\% \text{ of } 59.99\% \text{ of } 2499.99 - (2/5) \text{ of } 399.99 = ?$$

- a. 1020
- b. 1040
- c. 1060
- d. 1080
- e. 1000

30. Questions

$$(\sqrt{323.99} * 9.99) \div (\sqrt{399.99} \div \sqrt{0.99}) = ?^2 - 1.99^4$$

- a. 4
- b. 3
- c. 5

d. 6

e. 7

31. Questions

What value will come in the place of (?) in the following number series?

?, 56, 168, 420, 840, 1260

a. 12

b. 14

c. 16

d. 18

e. 20

32. Questions

20, 42, 26, 37, 32, ?

a. 35

b. 32

c. 37

d. 38

e. 39

33. Questions

15, 31, 95, 239, 495, ?

a. 875

b. 885

c. 895

d. 905

e. 915

34. Questions

8, ?, 24, 78, 320, 1610

a. 9

b. 10

c. 11

d. 12

e. 14

35. Questions

50, 61, ?, 91, 110, 133

a. 71

b. 72

c. 73

d. 74

e. 75

36. Questions

Find the wrong term in the following given series.

12, 16, 24, 51, 115, 240

a. 16

b. 24

c. 51

d. 12

e. 115

37. Questions

16, 10, 11, 19.5, 43, 112.5

a. 16

b. 10

c. 11

d. 19.5

e. 43

38. Questions

100, 112, 132, 162, 208, 260

a. 112

b. 132

c. 162

d. 208

e. 260

39. Questions

15, 23, 39, 73, 135, 263

a. 23

b. 39

c. 73

d. 135

e. 263

40. Questions

10, 26, 51, 89, 136, 200

a. 26

b. 51

c. 200

d. 136

e. 89

41. Questions

In each of the following questions, two equations are given. You have to solve both equations to find the relation between x and y.

I). $x^2 + 26x + 165 = 0$

II). $y^2 + 24y + 143 = 0$

a. $x > y$

b. $x \geq y$

c. $x < y$

d. $x \leq y$

e. $x = y$ or relationship can't be established

42. Questions

I). $3x + 2y = 42$

II). $2x + 3y = 43$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or relationship can't be established

43. Questions

I). $x^2 - 21x + 108 = 0$

II). $y^2 - 3y - 18 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or relationship can't be established

44. Questions

I). $x^2 - 11x + 28 = 0$

II). $y^2 - 5y + 4 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or relationship can't be established

45. Questions

I). $x^2 + 13x + 40 = 0$

II). $y^2 + 7y + 10 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$

e. $x = y$ or relationship can't be established

Explanations:

1. Questions

As per Note 1, the number of Petrol cars manufactured by company O is 120, and from the pie chart, the number of Petrol cars manufactured by company O is 10% of the total number of Petrol cars manufactured by all five companies together.

The total number of Petrol cars manufactured by all five companies together = $(120 / 10) \times 100 = 1200$

The number of Petrol cars manufactured by company L = 10% of 1200 = 120

The number of Petrol cars manufactured by company M = 50% of 1200 = 600

The number of Petrol cars manufactured by company N = 10% of 1200 = 120

The number of Petrol cars manufactured by company O = 10% of 1200 = 120

The number of Petrol cars manufactured by company P = 20% of 1200 = 240

As per Note 2, the total number of cars manufactured by company M is 300 more than the total number of cars manufactured by company P.

From the pie chart, the total number of cars manufactured by company M is 30% and the total number of cars manufactured by company P is 20% of the grand total, so the difference of 10% corresponds to 300 cars.

The total number of cars (Petrol + Diesel) manufactured by all five companies together = $(300 / 10) \times 100 = 3000$

The total number of cars manufactured by company L = 20% of 3000 = 600

The total number of cars manufactured by company M = 30% of 3000 = 900

The total number of cars manufactured by company N = 20% of 3000 = 600

The total number of cars manufactured by company O = 10% of 3000 = 300

The total number of cars manufactured by company P = 20% of 3000 = 600

The number of Diesel cars manufactured by each company = The total number of cars manufactured by that company – The number of Petrol cars manufactured by that company

The number of Diesel cars manufactured by company L = $600 - 120 = 480$

The number of Diesel cars manufactured by company M = $900 - 600 = 300$

The number of Diesel cars manufactured by company N = $600 - 120 = 480$

The number of Diesel cars manufactured by company O = $300 - 120 = 180$

The number of Diesel cars manufactured by company P = $600 - 240 = 360$

Final value table:

Companies	Number of Petrol Cars manufactured	Number of Diesel Cars manufactured	Total Number of Cars manufactured
L	120	480	600
M	600	300	900
N	120	480	600
O	120	180	300
P	240	360	600

Answer: B

The number of Diesel cars manufactured by company L = $600 - 120 = 480$

The number of Diesel cars manufactured by company M = $900 - 600 = 300$

The total number of Diesel cars manufactured by companies L and M together = $480 + 300 = 780$

The number of Diesel cars manufactured by company N = 480

Required percentage = $((780 - 480) / 480) * 100 = (300 / 480) * 100 = 62.5\%$

Hence, the correct answer is option B.

2. Questions

As per Note 1, the number of Petrol cars manufactured by company O is 120, and from the pie chart, the number of Petrol cars manufactured by company O is 10% of the total number of Petrol cars manufactured by all five companies together.

The total number of Petrol cars manufactured by all five companies together = $(120 / 10) \times 100 = 1200$

The number of Petrol cars manufactured by company L = 10% of 1200 = 120

The number of Petrol cars manufactured by company M = 50% of 1200 = 600

The number of Petrol cars manufactured by company N = 10% of 1200 = 120

The number of Petrol cars manufactured by company O = 10% of 1200 = 120

The number of Petrol cars manufactured by company P = 20% of 1200 = 240

As per Note 2, the total number of cars manufactured by company M is 300 more than the total number of cars manufactured by company P.

From the pie chart, the total number of cars manufactured by company M is 30% and the total number of cars manufactured by company P is 20% of the grand total, so the difference of 10% corresponds to 300 cars.

The total number of cars (Petrol + Diesel) manufactured by all five companies together = $(300 / 10) \times 100$
= 3000

The total number of cars manufactured by company L = 20% of 3000 = 600

The total number of cars manufactured by company M = 30% of 3000 = 900

The total number of cars manufactured by company N = 20% of 3000 = 600

The total number of cars manufactured by company O = 10% of 3000 = 300

The total number of cars manufactured by company P = 20% of 3000 = 600

The number of Diesel cars manufactured by each company = The total number of cars manufactured by that company – The number of Petrol cars manufactured by that company

The number of Diesel cars manufactured by company L = 600 – 120 = 480

The number of Diesel cars manufactured by company M = 900 – 600 = 300

The number of Diesel cars manufactured by company N = 600 – 120 = 480

The number of Diesel cars manufactured by company O = 300 – 120 = 180

The number of Diesel cars manufactured by company P = 600 – 240 = 360

Final value table:

Companies	Number of Petrol Cars manufactured	Number of Diesel Cars manufactured	Total Number of Cars manufactured
L	120	480	600
M	600	300	900
N	120	480	600
O	120	180	300
P	240	360	600

Answer: C

The number of Petrol cars manufactured by company M = 600

The number of Petrol cars manufactured by company A = 600 + (40% of 600) = 600 + 240 = 840

The number of Diesel cars manufactured by company O = 180

The number of Diesel cars manufactured by company A = 180 – (20% of 180) = 180 – 36 = 144

The total number of cars manufactured by company A = 840 + 144 = 984

Hence, the correct answer is option C.

3. Questions

As per Note 1, the number of Petrol cars manufactured by company O is 120, and from the pie chart, the number of Petrol cars manufactured by company O is 10% of the total number of Petrol cars manufactured by all five companies together.

The total number of Petrol cars manufactured by all five companies together = $(120 / 10) \times 100 = 1200$

The number of Petrol cars manufactured by company L = 10% of 1200 = 120

The number of Petrol cars manufactured by company M = 50% of 1200 = 600

The number of Petrol cars manufactured by company N = 10% of 1200 = 120

The number of Petrol cars manufactured by company O = 10% of 1200 = 120

The number of Petrol cars manufactured by company P = 20% of 1200 = 240

As per Note 2, the total number of cars manufactured by company M is 300 more than the total number of cars manufactured by company P.

From the pie chart, the total number of cars manufactured by company M is 30% and the total number of cars manufactured by company P is 20% of the grand total, so the difference of 10% corresponds to 300 cars.

The total number of cars (Petrol + Diesel) manufactured by all five companies together = $(300 / 10) \times 100 = 3000$

The total number of cars manufactured by company L = 20% of 3000 = 600

The total number of cars manufactured by company M = 30% of 3000 = 900

The total number of cars manufactured by company N = 20% of 3000 = 600

The total number of cars manufactured by company O = 10% of 3000 = 300

The total number of cars manufactured by company P = 20% of 3000 = 600

The number of Diesel cars manufactured by each company = The total number of cars manufactured by that company – The number of Petrol cars manufactured by that company

The number of Diesel cars manufactured by company L = $600 - 120 = 480$

The number of Diesel cars manufactured by company M = $900 - 600 = 300$

The number of Diesel cars manufactured by company N = $600 - 120 = 480$

The number of Diesel cars manufactured by company O = $300 - 120 = 180$

The number of Diesel cars manufactured by company P = $600 - 240 = 360$

Final value table:

Companies	Number of Petrol Cars manufactured	Number of Diesel Cars manufactured	Total Number of Cars manufactured
L	120	480	600
M	600	300	900
N	120	480	600
O	120	180	300
P	240	360	600

Answer: E

The number of Petrol cars manufactured by company L = 120

The number of Petrol cars manufactured by company M = 600

The number of Petrol cars manufactured by company N = 120

Required total = $120 + 600 + 120 = 840$

Hence, the correct answer is option E.

4. Questions

As per Note 1, the number of Petrol cars manufactured by company O is 120, and from the pie chart, the number of Petrol cars manufactured by company O is 10% of the total number of Petrol cars manufactured by all five companies together.

The total number of Petrol cars manufactured by all five companies together = $(120 / 10) \times 100 = 1200$

The number of Petrol cars manufactured by company L = 10% of 1200 = 120

The number of Petrol cars manufactured by company M = 50% of 1200 = 600

The number of Petrol cars manufactured by company N = 10% of 1200 = 120

The number of Petrol cars manufactured by company O = 10% of 1200 = 120

The number of Petrol cars manufactured by company P = 20% of 1200 = 240

As per Note 2, the total number of cars manufactured by company M is 300 more than the total number of cars manufactured by company P.

From the pie chart, the total number of cars manufactured by company M is 30% and the total number of cars manufactured by company P is 20% of the grand total, so the difference of 10% corresponds to 300 cars.

The total number of cars (Petrol + Diesel) manufactured by all five companies together = $(300 / 10) \times 100 = 3000$

The total number of cars manufactured by company L = 20% of 3000 = 600

The total number of cars manufactured by company M = 30% of 3000 = 900

The total number of cars manufactured by company N = 20% of 3000 = 600

The total number of cars manufactured by company O = 10% of 3000 = 300

The total number of cars manufactured by company P = 20% of 3000 = 600

The number of Diesel cars manufactured by each company = The total number of cars manufactured by that company – The number of Petrol cars manufactured by that company

The number of Diesel cars manufactured by company L = 600 – 120 = 480

The number of Diesel cars manufactured by company M = 900 – 600 = 300

The number of Diesel cars manufactured by company N = 600 – 120 = 480

The number of Diesel cars manufactured by company O = 300 – 120 = 180

The number of Diesel cars manufactured by company P = 600 – 240 = 360

Final value table:

Companies	Number of Petrol Cars manufactured	Number of Diesel Cars manufactured	Total Number of Cars manufactured
L	120	480	600
M	600	300	900
N	120	480	600
O	120	180	300
P	240	360	600

Answer: C

The number of Petrol cars manufactured by companies M, N and O together = 600 + 120 + 120 = 840

The total number of Diesel cars manufactured by companies L, O and P together = 480 + 180 + 360 = 1020

Required ratio = 840 : 1020 = 14 : 17

Hence, the correct answer is option C.

5. Questions

As per Note 1, the number of Petrol cars manufactured by company O is 120, and from the pie chart, the number of Petrol cars manufactured by company O is 10% of the total number of Petrol cars

manufactured by all five companies together.

The total number of Petrol cars manufactured by all five companies together = $(120 / 10) \times 100 = 1200$

The number of Petrol cars manufactured by company L = 10% of 1200 = 120

The number of Petrol cars manufactured by company M = 50% of 1200 = 600

The number of Petrol cars manufactured by company N = 10% of 1200 = 120

The number of Petrol cars manufactured by company O = 10% of 1200 = 120

The number of Petrol cars manufactured by company P = 20% of 1200 = 240

As per Note 2, the total number of cars manufactured by company M is 300 more than the total number of cars manufactured by company P.

From the pie chart, the total number of cars manufactured by company M is 30% and the total number of cars manufactured by company P is 20% of the grand total, so the difference of 10% corresponds to 300 cars.

The total number of cars (Petrol + Diesel) manufactured by all five companies together = $(300 / 10) \times 100 = 3000$

The total number of cars manufactured by company L = 20% of 3000 = 600

The total number of cars manufactured by company M = 30% of 3000 = 900

The total number of cars manufactured by company N = 20% of 3000 = 600

The total number of cars manufactured by company O = 10% of 3000 = 300

The total number of cars manufactured by company P = 20% of 3000 = 600

The number of Diesel cars manufactured by each company = The total number of cars manufactured by that company – The number of Petrol cars manufactured by that company

The number of Diesel cars manufactured by company L = $600 - 120 = 480$

The number of Diesel cars manufactured by company M = $900 - 600 = 300$

The number of Diesel cars manufactured by company N = $600 - 120 = 480$

The number of Diesel cars manufactured by company O = $300 - 120 = 180$

The number of Diesel cars manufactured by company P = $600 - 240 = 360$

Final value table:

Companies	Number of Petrol Cars manufactured	Number of Diesel Cars manufactured	Total Number of Cars manufactured
L	120	480	600
M	600	300	900
N	120	480	600
O	120	180	300
P	240	360	600

Answer: B

The number of Diesel cars manufactured by company M = 300

The total number of cars manufactured by company M = 900

Required percentage = $(300 / 900) * 100 = 33.33\%$

Hence, the correct answer is option B.

6. Questions

The number of wall clocks produced by Company B in each month = $(2 \times \text{The average number of wall clocks produced by A and B}) - \text{The number of wall clocks produced by Company A}$

The number of wall clocks produced by Company B in January = $(2 \times 150) - 120 = 180$

The number of wall clocks produced by Company B in February = $(2 \times 160) - 140 = 180$

The number of wall clocks produced by Company B in March = $(2 \times 175) - 195 = 155$

The number of wall clocks produced by Company B in April = $(2 \times 140) - 160 = 120$

The number of wall clocks produced by Company B in May = $(2 \times 170) - 150 = 190$

Final Table:

Months	The number of wall clocks produced by Company A	The number of wall clocks produced by Company B	Total number of wall clocks produced by Companies A and B together
January	120	180	300
February	140	180	320
March	195	155	350
April	160	120	280
May	150	190	340

Answer: B

The total number of wall clocks produced by Company A and Company B together in January = 300

The total number of wall clocks produced by Company A and Company B together in April = 280

Required ratio = $300 : 280 = 15 : 14$

Hence, the correct answer is option B.

7. Questions

The number of wall clocks produced by Company B in each month = $(2 \times \text{The average number of wall clocks produced by A and B}) - \text{The number of wall clocks produced by Company A}$

The number of wall clocks produced by Company B in January = $(2 \times 150) - 120 = 180$

The number of wall clocks produced by Company B in February = $(2 \times 160) - 140 = 180$

The number of wall clocks produced by Company B in March = $(2 \times 175) - 195 = 155$

The number of wall clocks produced by Company B in April = $(2 \times 140) - 160 = 120$

The number of wall clocks produced by Company B in May = $(2 \times 170) - 150 = 190$

Final Table:

Months	The number of wall clocks produced by Company A	The number of wall clocks produced by Company B	Total number of wall clocks produced by Companies A and B together
January	120	180	300
February	140	180	320
March	195	155	350
April	160	120	280
May	150	190	340

Answer: D

The total number of wall clocks produced by Companies A and B together in June = $280 + (50\% \text{ of } 280) = 280 + 140 = 420$

The number of wall clocks produced by Company B = $(2 / 7) * 420 = 120$

Hence, the correct answer is option D.

8. Questions

The number of wall clocks produced by Company B in each month = $(2 \times \text{The average number of wall clocks produced by A and B}) - \text{The number of wall clocks produced by Company A}$

The number of wall clocks produced by Company B in January = $(2 \times 150) - 120 = 180$

The number of wall clocks produced by Company B in February = $(2 \times 160) - 140 = 180$

The number of wall clocks produced by Company B in March = $(2 \times 175) - 195 = 155$

The number of wall clocks produced by Company B in April = $(2 \times 140) - 160 = 120$

The number of wall clocks produced by Company B in May = $(2 \times 170) - 150 = 190$

Final Table:

Months	The number of wall clocks produced by Company A	The number of wall clocks produced by Company B	Total number of wall clocks produced by Companies A and B together
January	120	180	300
February	140	180	320
March	195	155	350
April	160	120	280
May	150	190	340

Answer: B

The total number of wall clocks produced by both companies together in May = 340

The total number of wall clocks produced by both companies together in February = 320

Required percentage = $((340 - 320) / 320) * 100 = (20 / 320) * 100 = 6.25\%$

Hence, the correct answer is option B.

9. Questions

The number of wall clocks produced by Company B in each month = $(2 \times \text{The average number of wall clocks produced by A and B}) - \text{The number of wall clocks produced by Company A}$

The number of wall clocks produced by Company B in January = $(2 \times 150) - 120 = 180$

The number of wall clocks produced by Company B in February = $(2 \times 160) - 140 = 180$

The number of wall clocks produced by Company B in March = $(2 \times 175) - 195 = 155$

The number of wall clocks produced by Company B in April = $(2 \times 140) - 160 = 120$

The number of wall clocks produced by Company B in May = $(2 \times 170) - 150 = 190$

Final Table:

Months	The number of wall clocks produced by Company A	The number of wall clocks produced by Company B	Total number of wall clocks produced by Companies A and B together
January	120	180	300
February	140	180	320
March	195	155	350
April	160	120	280
May	150	190	340

Answer: A

The number of wall clocks produced by Company A in May = 150

The number of wall clocks produced by Company A in June = $150 + (30\% \text{ of } 150) = 150 + 45 = 195$

The number of wall clocks produced by Company B in April = 120

The number of wall clocks produced by Company B in June = $120 + (20\% \text{ of } 120) = 120 + 24 = 144$

The total number of wall clocks produced by both companies in June = $195 + 144 = 339$

Hence, the correct answer is option A.

10. Questions

The number of wall clocks produced by Company B in each month = $(2 \times \text{The average number of wall clocks produced by A and B}) - \text{The number of wall clocks produced by Company A}$

The number of wall clocks produced by Company B in January = $(2 \times 150) - 120 = 180$

The number of wall clocks produced by Company B in February = $(2 \times 160) - 140 = 180$

The number of wall clocks produced by Company B in March = $(2 \times 175) - 195 = 155$

The number of wall clocks produced by Company B in April = $(2 \times 140) - 160 = 120$

The number of wall clocks produced by Company B in May = $(2 \times 170) - 150 = 190$

Final Table:

Months	The number of wall clocks produced by Company A	The number of wall clocks produced by Company B	Total number of wall clocks produced by Companies A and B together
January	120	180	300
February	140	180	320
March	195	155	350
April	160	120	280
May	150	190	340

Answer: C

The number of wall clocks produced by Company A in January = 120

The number of wall clocks produced by Company A in February = 140

The number of wall clocks produced by Company A in March = 195

The number of wall clocks produced by Company A in April = 160

Required total = $120 + 140 + 195 + 160 = 615$

Hence, the correct answer is option C.

11. Questions

Answer: C

Quantity of liquid A left = initial quantity * $(1 - \text{quantity drawn each time} / \text{initial quantity})^3$

Quantity of liquid A left = $250 * (1 - 25/250)^3$

Quantity of liquid A left = $250 * (9/10)^3$

Quantity of liquid A left = $250 * (729/1000)$

Quantity of liquid A left = 182.25 L

Hence, the correct answer is option C.

12. Questions

Answer: D

The ratio of the present ages of Kesav and Kumar = $3 : 5 = 6 : 10$

The ratio of the present ages of Kesav and Kamal = $6 : 7$

The ratio of the present ages of Kesav, Kumar and Kamal = 6 : 10 : 7

Let their present ages be $6u$, $10u$ and $7u$ years respectively.

Average of the present ages of Kumar and Kamal = $(10u + 7u)/2 = 34$

$$17u = 68$$

$$u = 4$$

Present age of Kesav = $6 * 4 = 24$ years

Age of Kesav after 5 years = $24 + 5 = 29$ years

Hence, the correct answer is option D.

13. Questions

Answer: C

Let the total distance of the journey be $6D$ km.

Distance covered in the first stage = D

Remaining distance = $6D - D = 5D$

Distance covered in the second stage = $(1/2) * 5D = 2.5D$

Remaining distance = $5D - 2.5D = 2.5D$

This remaining distance is covered in the third and fourth stages in the ratio 4 : 1.

Distance covered in the fourth stage = $(1/5) * 2.5D = 0.5D$

$$2.5D - 0.5D = 120$$

$$2D = 120$$

$$D = 60$$

Total distance of the journey $6D = 6 * 60 = 360$ km

Distance covered in the first stage = $D = 60$ km

Time taken in the first stage = $60/30 = 2$ hours

Hence, the correct answer is option C.

14. Questions

Answer: A

Total number of balls in the bag = $8 + 7 + 9 = 24$

Number of red balls = 8

Number of yellow balls = 7

Number of favourable outcomes = $8 + 7 = 15$

Required probability = $15/24 = 5/8$

Hence, the correct answer is option A.

15. Questions

Answer: A

Selling price of the article after discount = Rs.1584

Marked price = $1584 * (100/88) = \text{Rs.}1800$

Cost price = $1584 * (100/120) = \text{Rs.}1320$

If sold at the marked price, profit = $1800 - 1320 = \text{Rs.}480$

Profit percentage = $(480/1320) * 100 = 36.36\%$

Hence, the correct answer is option A.

16. Questions

Answer: D

P completes one-fourth of the work in 3 days, so P alone completes the whole work in 12 days.

Q completes one-third of the work in 6 days, so Q alone completes the whole work in 18 days.

R completes half of the work in 18 days, so R alone completes the whole work in 36 days.

Let the total work be 36 units.

The efficiency of P = $36/12 = 3$ units

The efficiency of Q = $36/18 = 2$ units

The efficiency of R = $36/36 = 1$ unit

One day work of all three together = $3 + 2 + 1 = 6$ units

Work done by Q alone in the last 6 days = $2 * 6 = 12$ units

$$6n + 12 = 36$$

$$6n = 24$$

$$n = 4$$

Hence, the correct answer is option D.

17. Questions

Answer: B

Amount at the end of 2 years = Rs.10000

Amount at the end of 3 years = Rs.12500

Interest earned in the third year = $12500 - 10000 = \text{Rs.}2500$

This interest is earned on the amount at the end of 2 years.

Rate of interest = $(2500/10000) * 100 = 25\%$

Let the sum invested be Rs.P.

$$P * (125/100)^2 = 10000$$

$$P * (25/16) = 10000$$

$$P = 10000 * (16/25)$$

$$P = \text{Rs.}6400$$

Hence, the correct answer is option B.

18. Questions

Answer: D

Let the side of the square be $5x$ cm and the radius of the semicircle be $7x$ cm.

$$\text{Perimeter of the square} = 4 * 5x = 20x$$

$$\text{Perimeter of the semicircle} = \pi r + d = (22/7) * 7x + 2 * 7x$$

$$\text{Perimeter of the semicircle} = 22x + 14x = 36x$$

$$20x + 36x = 280$$

$$56x = 280$$

$$x = 5$$

$$\text{Radius of the semicircle} = 7 * 5 = 35 \text{ cm}$$

$$\text{Area of the semicircle} = (1/2) * (22/7) * 35^2$$

$$\text{Area of the semicircle} = (1/2) * (22/7) * 1225$$

$$\text{Area of the semicircle} = 1925 \text{ cm}^2$$

Hence, the correct answer is option D.

19. Questions

Answer: D

Let the speed of the stream be s km/hr.

$$\text{Upstream speed} = (13 - s) \text{ km/hr}$$

$$\text{Downstream speed} = (13 + s) \text{ km/hr}$$

$$60/(13 - s) + 96/(13 + s) = 12$$

$$5/(13 - s) + 8/(13 + s) = 1$$

$$65 + 5s + 104 - 8s = 169 - s^2$$

$$169 - 3s = 169 - s^2$$

$$s^2 = 3s$$

$$s = 3 \text{ km/hr}$$

$$\text{Upstream speed} = 13 - 3 = 10 \text{ km/hr}$$

$$\text{Downstream speed} = 13 + 3 = 16 \text{ km/hr}$$

$$\text{Time taken to cover 45 km upstream} = 45/10 = 4.5 \text{ hours}$$

$$4.5 \text{ hours} = 4 \text{ hours } 30 \text{ minutes}$$

Hence, the correct answer is option D.

20. Questions

Answer: C

$$\text{Investment of A for the whole year} = 3600 * 12 = 43200$$

B invested Rs.4800 for the first 4 months.

$$\text{After the withdrawal, the investment of B} = 4800 - 1200 = \text{Rs.}3600$$

B invested Rs.3600 for the remaining 8 months.

$$\text{Investment of B} = 4800 * 4 + 3600 * 8$$

$$\text{Investment of B} = 19200 + 28800 = 48000$$

$$\text{Ratio of the shares of A and B} = 43200 : 48000 = 9 : 10$$

$$\text{Total number of parts} = 9 + 10 = 19$$

$$\text{Value of one part} = 28500/19 = 1500$$

$$\text{Share of B} = 10 * 1500 = \text{Rs.}15000$$

Hence, the correct answer is option C.

21. Questions

Answer: C

$$\Rightarrow 16 * 8 + 40\% \text{ of } 1500 = ? + 68$$

$$\Rightarrow 128 + 40\% \text{ of } 1500 = ? + 68$$

$$\Rightarrow 128 + 600 = ? + 68$$

$$\Rightarrow 728 = ? + 68$$

$$\Rightarrow ? = 728 - 68$$

$$\Rightarrow ? = 660$$

Hence, the correct answer is option C.

22. Questions

Answer: B

$$\Rightarrow 47 + 15 * \frac{6}{25} - 12.6 = ? * \frac{16}{4}$$

$$\Rightarrow 47 + 3.6 - 12.6 = ? * 4$$

$$\Rightarrow 50.6 - 12.6 = ? * 4$$

$$\Rightarrow 38 = ? * 4$$

$$\Rightarrow ? = 38 / 4$$

$$\Rightarrow ? = 9.5$$

Hence, the correct answer is option B.

23. Questions

Answer: C

$$\Rightarrow \sqrt{[12 * 12 \div (1/9)]} + 18 * 15 = ? * 18$$

$$\Rightarrow \sqrt{(144 * 9)} + 270 = ? * 18$$

$$\Rightarrow \sqrt{1296} + 270 = ? * 18$$

$$\Rightarrow 36 + 270 = ? * 18$$

$$\Rightarrow 306 = ? * 18$$

$$\Rightarrow ? = 306 / 18$$

$$\Rightarrow ? = 17$$

Hence, the correct answer is option C.

24. Questions

Answer: E

$$\Rightarrow 35 * \frac{24}{56} + \frac{84}{3} - \frac{77}{5} = ?$$

$$\Rightarrow 15 + 28 - 15.4 = ?$$

$$\Rightarrow 43 - 15.4 = ?$$

$$\Rightarrow ? = 27.6$$

Hence, the correct answer is option E.

25. Questions

Answer: A

$$\Rightarrow ? \div 5 + 144 \div 2.4 * 2 = 14 * 15$$

$$\Rightarrow ? \div 5 + 60 * 2 = 210$$

$$\Rightarrow ? \div 5 + 120 = 210$$

$$\Rightarrow ? \div 5 = 210 - 120$$

$$\Rightarrow ? \div 5 = 90$$

$$\Rightarrow ? = 90 * 5$$

$$\Rightarrow ? = 450$$

Hence, the correct answer is option A.

26. Questions

Answer: C

$$\Rightarrow \sqrt{399.99} * 17.99 \div 5.98 = ?^2 + (63.99)^{1/3} * 6.01$$

$$\Rightarrow \sqrt{400} * 18 \div 6 = ?^2 + (64)^{1/3} * 6$$

$$\Rightarrow 20 * (18 / 6) = ?^2 + 4 * 6$$

$$\Rightarrow 20 * 3 = ?^2 + 24$$

$$\Rightarrow 60 = ?^2 + 24$$

$$\Rightarrow ?^2 = 60 - 24$$

$$\Rightarrow ?^2 = 36$$

$$\Rightarrow ? = 6$$

Hence, the correct answer is option C.

27. Questions

Answer: D

$$\Rightarrow 149.99\% \text{ of } 199.98 - 359.99 \div \sqrt{143.99} = \sqrt{323.99} * ?$$

$$\Rightarrow 150\% \text{ of } 200 - 360 \div \sqrt{144} = \sqrt{324} * ?$$

$$\Rightarrow (150/100) * 200 - 360 \div 12 = 18 * ?$$

$$\Rightarrow 300 - 30 = 18 * ?$$

$$\Rightarrow 270 = 18 * ?$$

$$\Rightarrow ? = 270 / 18$$

$$\Rightarrow ? = 15$$

Hence, the correct answer is option D.

28. Questions

Answer: A

$$\Rightarrow (24.99)^2 + (19.98)^2 - (29.99)^2 + \sqrt{255.99} = ?$$

$$\Rightarrow 25^2 + 20^2 - 30^2 + \sqrt{256} = ?$$

$$\Rightarrow 625 + 400 - 900 + 16 = ?$$

$$\Rightarrow 1025 - 900 + 16 = ?$$

$$\Rightarrow 125 + 16 = ?$$

$$\Rightarrow ? = 141$$

Hence, the correct answer is option A.

29. Questions

Answer: B

$$\Rightarrow 79.98\% \text{ of } 59.99\% \text{ of } 2499.99 - (2/5) \text{ of } 399.99 = ?$$

$$\Rightarrow 80\% \text{ of } 60\% \text{ of } 2500 - (2/5) * 400 = ?$$

$$\Rightarrow (80/100) * (60/100) * 2500 - 2 * 80 = ?$$

$$\Rightarrow 4/5 * 60 * 25 - 160 = ?$$

$$\Rightarrow 48 * 25 - 160 = ?$$

$$\Rightarrow 1200 - 160 = ?$$

$$\Rightarrow ? = 1040$$

Hence, the correct answer is option B.

30. Questions

Answer: C

$$\Rightarrow (\sqrt{323.99} * 9.99) \div (\sqrt{399.99} \div \sqrt{0.99}) = ?^2 - 1.99^4$$

$$\Rightarrow (\sqrt{324} * 10) \div (\sqrt{400} \div \sqrt{1}) = ?^2 - 2^4$$

$$\Rightarrow (18 * 10) \div (20 \div 1) = ?^2 - 16$$

$$\Rightarrow 180 \div 20 = ?^2 - 16$$

$$\Rightarrow 9 = ?^2 - 16$$

$$\Rightarrow ?^2 = 9 + 16$$

$$\Rightarrow ?^2 = 25$$

$$\Rightarrow ? = 5$$

Hence, the correct answer is option C.

31. Questions

Answer: C

The series follows the pattern of multiplying numbers, which are decreasing 0.5 starting from 3.5:

$$? * 3.5 = 56 \Rightarrow ? = 16$$

$$56 * 3 = 168$$

$$168 * 2.5 = 420$$

$$420 * 2 = 840$$

$$840 * 1.5 = 1260$$

$$\text{So, } ? = 16.$$

Hence, the correct answer is option C.

32. Questions

Answer: B

The series consists of two alternate series:

1st, 3rd, 5th terms: 20, 26, 32 (Pattern: +6)

2nd, 4th, 6th terms: 42, 37, ? (Pattern: -5)

$$42 - 5 = 37$$

$$37 - 5 = 32$$

$$\text{So, } ? = 32.$$

Hence, the correct answer is option B.

33. Questions

Answer: C

The series follows the pattern of adding squares of consecutive multiples of 4:

$$15 + 4^2 = 31$$

$$31 + 8^2 = 95$$

$$95 + 12^2 = 239$$

$$239 + 16^2 = 495$$

$$495 + 20^2 = 895$$

$$\text{So, } ? = 895.$$

Hence, the correct answer is option C.

34. Questions

Answer: B

The series follows the pattern of $(* n + 2n)$ where $n = 1, 2, 3, 4, 5$:

$$8 * 1 + 2 = 10$$

$$10 * 2 + 4 = 24$$

$$24 * 3 + 6 = 78$$

$$78 * 4 + 8 = 320$$

$$320 * 5 + 10 = 1610$$

$$\text{So, ?} = 10.$$

Hence, the correct answer is option B.

35. Questions

Answer: D

The series follows the pattern of adding consecutive prime numbers (+11, +13, +17, +19, +23):

$$50 + 11 = 61$$

$$61 + 13 = 74$$

$$74 + 17 = 91$$

$$91 + 19 = 110$$

$$110 + 23 = 133$$

$$\text{So, ?} = 74.$$

Hence, the correct answer is option D.

36. Questions

Answer: D

$$15 + 1^3 = 16$$

$$16 + 2^3 = 24$$

$$24 + 3^3 = 51$$

$$51 + 4^3 = 115$$

$$115 + 5^3 = 240$$

Hence, the correct answer is option D.

37. Questions

Answer: B

$$16 * 0.5 + 1 = 9 \text{ (Therefore, 10 is wrong and should be 9)}$$

$$9 * 1 + 2 = 11$$

$$11 * 1.5 + 3 = 19.5$$

$$19.5 * 2 + 4 = 43$$

$$43 * 2.5 + 5 = 112.5$$

Logic: Pattern is $*0.5 + 1$, $*1 + 2$, $*1.5 + 3$, $*2 + 4$, $*2.5 + 5$.

Hence, the correct answer is option B.

38. Questions

Answer: D

$$100 + 12 = 112$$

$$112 + 20 = 132$$

$$132 + 30 = 162$$

$$162 + 42 = \mathbf{204}$$
 (Therefore, 208 is wrong and should be 204)

$$204 + 56 = 260$$

Logic: Consecutive differences are 12, 20, 30, 42, 56 with a double difference of +8, +10, +12, +14.

Hence, the correct answer is option D.

39. Questions

Answer: C

$$15 + 8 = 23$$

$$23 + 16 = 39$$

$$39 + 32 = \mathbf{71}$$
 (Therefore, 73 is wrong and should be 71)

$$71 + 64 = 135$$

$$135 + 128 = 263$$

Logic: The difference between consecutive terms doubles each time (8, 16, 32, 64, 128).

Hence, the correct answer is option C.

40. Questions

Answer: E

$$10 + 4^2 = 26$$

$$26 + 5^2 = 51$$

$$51 + 6^2 = \mathbf{87}$$
 (Therefore, 89 is wrong and should be 87)

$$87 + 7^2 = 136$$

$$136 + 8^2 = 200$$

Logic: The difference between consecutive terms is the square of consecutive numbers starting from 4.

Hence, the correct answer is option E.

41. Questions

Answer: E

I). $x^2 + 26x + 165 = 0$

$$x^2 + 11x + 15x + 165 = 0$$

$$x(x + 11) + 15(x + 11) = 0$$

$$(x + 11)(x + 15) = 0$$

$$x = -11, -15$$

II). $y^2 + 24y + 143 = 0$

$$y^2 + 11y + 13y + 143 = 0$$

$$y(y + 11) + 13(y + 11) = 0$$

$$(y + 11)(y + 13) = 0$$

$$y = -11, -13$$

Comparison:

$$-11 = -11$$

$$-11 > -13$$

$$-15 < -11$$

$$-15 < -13$$

$x = y$ or relationship can't be established.

Hence, the correct answer is option E.

42. Questions

Answer: C

I). $3x + 2y = 42$

II). $2x + 3y = 43$

Multiply equation I by 2 and II by 3:

$$6x + 4y = 84$$

$$6x + 9y = 129$$

Subtracting the equations:

$$5y = 45$$

$$y = 9$$

Substituting the value of y in equation I:

$$3x + 2(9) = 42$$

$$3x + 18 = 42$$

$$3x = 24$$

$$x = 8$$

Comparison:

$$8 < 9$$

$$x < y$$

Hence, the correct answer is option C.

43. Questions

Answer: A

I). $x^2 - 21x + 108 = 0$

$$x^2 - 12x - 9x + 108 = 0$$

$$x(x - 12) - 9(x - 12) = 0$$

$$(x-12)(x-9) = 0$$

$$x = 12, 9$$

II). $y^2 - 3y - 18 = 0$

$$y^2 - 6y + 3y - 18 = 0$$

$$y(y - 6) + 3(y - 6) = 0$$

$$(y-6)(y+3) = 0$$

$$y = 6, -3$$

Comparison:

$$12 > 6$$

$$12 > -3$$

$$9 > 6$$

$$9 > -3$$

$$x > y$$

Hence, the correct answer is option A.

44. Questions

Answer: B

I). $x^2 - 11x + 28 = 0$

$$x^2 - 7x - 4x + 28 = 0$$

$$x(x - 7) - 4(x - 7) = 0$$

$$(x-7)(x-4) = 0$$

$$x = 7, 4$$

II). $y^2 - 5y + 4 = 0$

$$y^2 - 4y - 1y + 4 = 0$$

$$y(y - 4) - 1(y - 4) = 0$$

$$(y-4)(y-1) = 0$$

$$y = 4, 1$$

Comparison:

$$7 > 4$$

$$7 > 1$$

$$4 = 4$$

$$4 > 1$$

$$x \geq y$$

Hence, the correct answer is option B.

45. Questions

Answer: D

I). $x^2 + 13x + 40 = 0$

$$x^2 + 8x + 5x + 40 = 0$$

$$x(x + 8) + 5(x + 8) = 0$$

$$(x+8)(x+5) = 0$$

$$x = -8, -5$$

II). $y^2 + 7y + 10 = 0$

$$y^2 + 5y + 2y + 10 = 0$$

$$y(y + 5) + 2(y + 5) = 0$$

$$(y+5)(y+2) = 0$$

$$y = -5, -2$$

Comparison:

$$-8 < -5$$

$$-8 < -2$$

$$-5 = -5$$

$$-5 < -2$$

$$x \leq y$$

Hence, the correct answer is option D.

@QuizByMickey